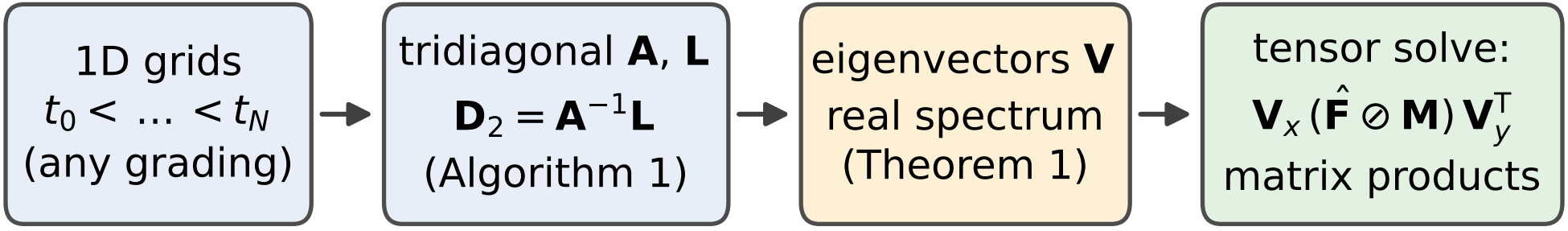

classical HWCM:
one dense $N^2 \times N^2$ system
$O(N^6)$ work

this work:
$O(N^3)$ work in 2D
on any graded grid